

IPD Tuning Guide

Power Application Controller™





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1 INTRODUCTION

This document describes the procedure for tuning the PAC52XX Initial Position Detection (IPD). The IPD implementation can be used for internal permanent magnet (IPM) motors, which exhibit magnetic saliency. The magnetic saliency results in the motor having L_d and L_q values that are not equal which is necessary for the IPD algorithm to work. For surface mounted permanent magnet (SPM) motors, $L_d = L_q$; therefore, this IPD algorithm will not produce acceptable results for SPM motors.

The IPD algorithm calculates an IPD Angle with a 15 degree resolution; however, the IPD Angle accuracy may be less depending on many factors related to the motor being tuned

2 IPD TUNING OVERVIEW

During IPD Tuning the following steps should be followed:

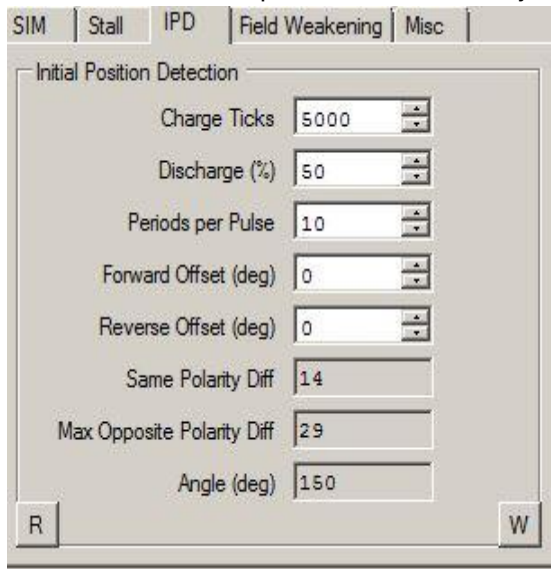
- Adjust Initial Charge Ticks
- Adjust Discharge Percent to ensure current pulses return to 0 amps.
- Adjust Charge Ticks for Accurate IPD Angle
- Adjust Periods per Pulse to reduce motor noise
- Adjust Forward/Reverse Angle Offsets for desired Open Loop startup
- Test IPD
- Tune Startup after IPD

3 IPD TUNING PARAMETERS

The following #defines in param_default.h can be used to store the IPD parameters after tuning with the GUI:

```
#define IPD_CHARGE_TICKS_DEFAULT_VALUE      300
#define IPD_DISCHARGE_PERCENT_DEFAULT_VALUE 85
#define IPD_PERIODS_PER_PULSE_DEFAULT_VALUE 1
#define IPD_ANGLE_FORWARD_OFFSET_DEFAULT_VALUE 0
#define IPD_ANGLE_REVERSE_OFFSET_DEFAULT_VALUE 0
```

To tune IPD, the IPD parameters can be adjusted from the GUI as indicated by the figure below.



4 IPD TUNING PROCEDURE

The IPD tuning procedure described assumes that the motor FOC parameters have been tuned for proper operation without IPD.

4.1 Initial Charge Ticks Adjustment

The IPD Charge Ticks need to be adjusted so the motor current pulses can be observed.

1. Start with the following set of parameters:

- Charge Ticks = 100
- Discharge Percent = 85%
- Periods per Pulse = 1
- Set Id ref and Iq ref = 0. This will keep the motor from moving after IPD
- Forward and Reverse offsets = 0

2. Add a current probe to one of the motor phases.

3. Enable the motor and observe the phase current pulses created by IPD using the oscilloscope.

4. If the current pulses aren't reaching at least 2.5% of the maximum motor current, increase the IPD Charge Ticks until they do achieve 2.5% of the maximum motor current. For some motors, the IPD Charge Ticks value may need to be 300, 3000, 12000, or even higher.

4.2 Adjust Discharge Percent

Adjusting the Discharge Percent ensures that the current pulses return close to 0 amps after each pulse.

1. Enable the motor to observe the current pulses.

- At the end of each pulse, the current should be close to 0 amps (See figure below).

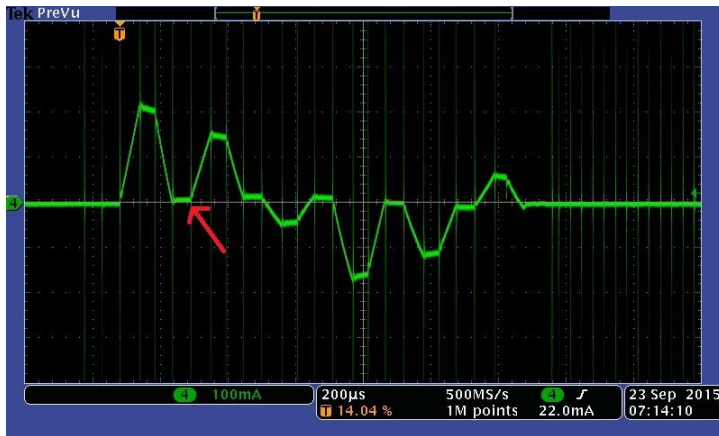


Figure 1: Correct Current Pulses

- If after each pulse, the current is not close to 0 amps, then adjustment of the IPD Discharge Percent is needed.
- If the current is overshooting 0 amps, then decrease the discharge percent (see figure below).

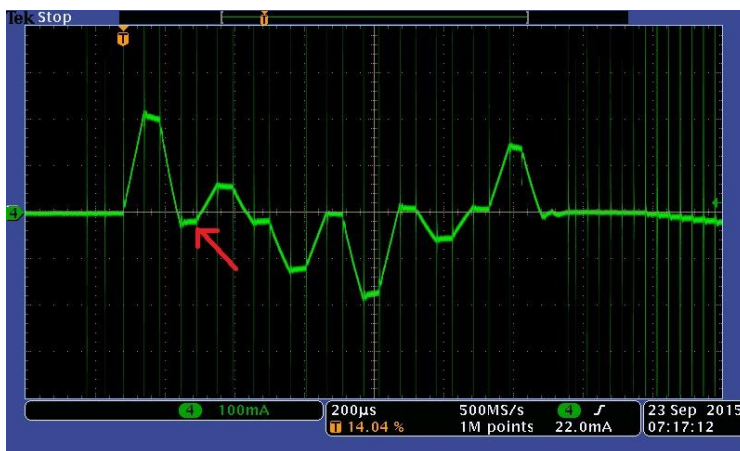


Figure 2: Current Pulses Overshoot

- If the current is not reaching close to 0 amps, then increase the discharge percent (see figure below).



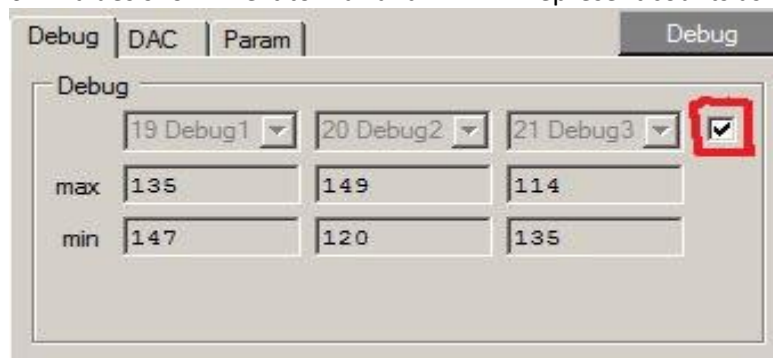
Figure 3: Current Pulses Undershoot

6. Revisit the Discharge Percent after adjusting the charge ticks in the next section to ensure the currents are returning close to 0 amps.

4.3 Adjust Charge Ticks for Accurate IPD Angle

Additional adjustment of the Charge Ticks ensures that the current pulses are large enough to calculate the IPD angle.

1. Use the GUI to set I_d ref and I_q ref = 0. This will keep the motor from moving after IPD.
2. Set the debug box to see the IPD current pulse measurements.
 - a. Change Debug selection to “19 Debug1, 20 Dubug2, 21 Debug3
 - b. Check the box to the right side of 21 Debug3 to enter special mode
 - c. Values shown next to max and min will represent counts as measured by the ADC



3. Enable the motor to let IPD run.
4. Disable the motor and view the currents measured in the debug window.
 - a. If the shunt resistors and current input gains are chosen properly, then a max current in the range of 80 counts will typically result in good IPD angle calculation. But some configurations and some motors will require much more than 80 counts for an accurate IPD angle.

5. Increase the Charge Ticks and repeat the procedure until the Same Polarity Current Difference (Same Polarity Diff) value and the Maximum to Opposite Current Difference (Max-Opposite Polarity Diff) are both consistently ≥ 5 counts. This will typically ensure that the current measurements provide enough separation to calculate a reasonably accurate IPD angle.
 - a. **Note that for some motors, these values may need to be consistently higher than 5 for correct calculation of the IPD Angle.**

4.4 Verify IPD Angle Calculation

This section describes the process to verify that the IPD Angle is being calculated with a reasonable accuracy.

1. Make sure the rotor has some mark on it that can be referred to as the rotor is moved.
2. Set I_{d_ref} and $I_{q_ref} = 0$ so the motor doesn't move after IPD
3. Enable the motor so that IPD is performed.
4. Note the IPD Angle on the GUI.
5. Rotate the motor clockwise slightly.
6. Repeat steps 2-4 noting the IPD Angle each time.
7. Ensure that the IPD Angle is increasing. It should move through angles between 0 and 360, then repeat 0-360 degrees as you continue to rotate the motor clockwise.
8. If the IPD Angle is not increasing appropriately as the motor is rotated clockwise, return to section 4.3 and increase the Charge Ticks so that larger current pulses are created.
 - a. This will help IPD calculate a more accurate angle.
 - b. **Note that for some motors the IPD angle calculation error may result in the angle decreasing slightly even though the motor has been rotated clockwise from the previous position. This is acceptable and can be compensated for by the offset angle settings.**

4.5 Modify Periods per Pulse to reduce noise

To minimize the noise generated by IPD, the IPD Charge Ticks should be as small as possible but large enough for the IPD angle to be calculated accurately enough for motor startup. If the IPD is being calculated accurately, but the noise is not acceptable, then adjusting Periods per Pulse can help reduce the noise.

1. Increment Periods per Pulse by 1
2. Repeat sections 4.2 and 4.4 to adjust the Discharge Percent, Charge Ticks, and verify the IPD angle accuracy.
 - a. Make sure that the current returns to near 0 Amps after each pulse
 - b. It may be necessary to count the charge periods and discharge periods to know at what point the current should be 0 Amps.
 - i. For example: If Periods per Pulse is 4, then for the first pulse, there should be 4 charge cycles followed by 4 discharge cycles. The current should be close to 0 amps at the end of the 4 discharge cycles.
3. Repeat steps 1 and 2 until the noise is reduced.

The Periods per Pulse for some motors may be 4, 16, 40, or even higher before the noise is acceptable. If the Periods per Pulse is too large, the motor may show movement during IPD.

4.6 Adjust Forward Offset

Before adjusting the forward or reverse offsets, make sure the Id ref and Iq ref values have been tuned for proper Open Loop operation. This is necessary because changes to Id/Iq ref may affect the Forward and Reverse angle offsets that are needed to make the startup procedure work best.

1. Disable Start In Motion (SIM) to speed up this process.
2. Ensure Id ref and Iq ref values are set for Open Loop operation.
3. Set Align time to 30 seconds.
4. Enable the motor. After IPD, the motor will align.
5. Observe the direction the motor moves.
6. If the motor moves backwards, then increase the forward offset angle and repeat steps 4-5 until it moves forward.
7. If the motor moves forward, then note the IPD Angle calculated. This is IPD Angle 1.
8. Disable the motor and let the rotor stop moving.
9. Enable the motor again.
10. Note the IPD Angle calculated, this is IPD Angle 2.
11. The Delta Angle is $\sim \text{IPD Angle 2} - \text{IPD Angle 1}$.
 - o The reason the Delta Angle is approximate is that motor cogging may cause the motor to move after alignment but before IPD Angle 2 is calculated.
12. Adjust the forward Offset to achieve a Delta Angle typically between 30 and 60 degrees.

The Delta Angle computed in step 11 should typically be between 30 and 60 electrical degrees to ensure correct motor rotation at startup. Offsets to achieve this Delta Angle are necessary to overcome errors in the IPD angle calculation and prevent the motor from moving in the wrong direction. For some motors, a larger Delta Angle may be necessary due to larger IPD angle calculation errors.

Note that the forward offset may be negative to achieve this and is dependent on the Id ref and Iq ref values used for Open Loop startup.

4.7 Adjust Reverse Offset

The reverse offset adjustment process is identical to the forward offset adjustment except the motor should move in reverse.

4.8 Test IPD

To test the IPD, perform the following:

1. Set the Align time to 0.0 seconds or a very small value like 0.1 seconds.
2. Enable the motor and verify that the motor moves in the correct direction.
3. Disable the motor.
4. Allow the rotor to stop moving.
5. Position the motor to a new angle and repeat steps 2-5.
6. Repeat the procedure over a complete 360 degree mechanical revolution of the motor to ensure that IPD is working properly.
7. If IPD causes the motor to move backwards for any position, then increase Charge Ticks so a more accurate IPD angle can be calculated and repeat sections 4.3 through 4.8.

4.9 Tuning the startup after IPD

Before tuning startup for IPD, make sure that the motor Open Loop startup is tuned properly for operation without IPD.



To tune the startup after IPD, the following parameters should be adjusted:

- Align Time
 - An align time is not necessary following IPD.
 - 0.0 seconds typically results in the smoothest startup
- Open Loop Speed Ramp –
 - It may be necessary to increase the OL speed ramp to get a smoother startup following IPD
- Id/Iq ramp rate

Additional instructions on how to adjust these parameters may be provided in the future.



CONTACT INFORMATION

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